

STUDENT RESPONSE SHEET - ANANYA SHARMA

ANSWER BOOKLET - DESCRIPTIVE GRADING

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Answer 1:

A stack is a linear data structure that follows the LIFO (Last In First Out) principle. Elements are inserted (push) and removed (pop) from the same end, called the top pointer. Real-world example: A stack of plates or backtracking in maze routing.

A queue is a linear data structure following FIFO (First In First Out). Insertion (enqueue) happens at the rear, and removal (dequeue) happens at the front pointer. Example: A queue of people waiting at a movie ticket counter.

Differences: Stack uses one end for insertion/deletion (LIFO), whereas queue uses two distinct ends (FIFO). Stacks are used in function calls, while queues are used in CPU scheduling.

Answer 2:

TCP (Transmission Control Protocol) is connection-oriented, reliable, and guarantees in-order delivery using a three-way handshake (SYN, SYN-ACK, ACK). It has flow control and error recovery. UDP (User Datagram Protocol) is connectionless, fast, and lightweight but does not guarantee packet delivery or order. TCP uses sliding window protocol; UDP just blasts packets.

TCP is used for Web (HTTP), Email (SMTP); UDP is used for live streaming, DNS, and online gaming. Diagram depicts TCP handshake client-server exchange and UDP single-packet dispatch.

Answer 3:

Binary search is an efficient search algorithm on sorted lists. It works by repeatedly dividing the search range in half. If the target value is less than the middle element, it narrows the search to the lower half; otherwise, it narrows to the upper half. Here is the implementation in Python:

```
def binary_search(arr, target):  
    low = 0  
    high = len(arr) - 1  
    while low <= high:  
        mid = (low + high) // 2  
        if arr[mid] == target: return mid  
        elif arr[mid] < target: low = mid + 1  
        else: high = mid - 1  
    return -1
```

Time complexity: Best case is $O(1)$ if target is at middle. Average and worst-case time complexity is $O(\log n)$ because the search space is divided by 2 at each step.

Answer 4:

Bayes' Theorem states: $P(A|B) = P(B|A) * P(A) / P(B)$. Let D = Patient has the disease, and Pos = Patient tests positive.

Given: Prevalence of disease, $P(D) = 0.1\% = 0.001$. Therefore, $P(\text{not } D) = 1 - 0.001 = 0.999$. Test accuracy for disease positive (sensitivity), $P(\text{Pos}|D) = 99\% = 0.99$. False positive rate (assuming test is 99% accurate overall, false positive is 1%), $P(\text{Pos}|\text{not } D) = 1\% = 0.01$. We need to find $P(D|\text{Pos})$.

Using Bayes' Formula: $P(D|\text{Pos}) = P(\text{Pos}|D) * P(D) / [P(\text{Pos}|D) * P(D) + P(\text{Pos}|\text{not } D) * P(\text{not } D)] = (0.99 * 0.001) / [(0.99 * 0.001) + (0.01 * 0.999)] = 0.00099 / [0.00099 + 0.00999] = 0.00099 / 0.01098 = 0.09016$, which is approximately 9.02% probability.